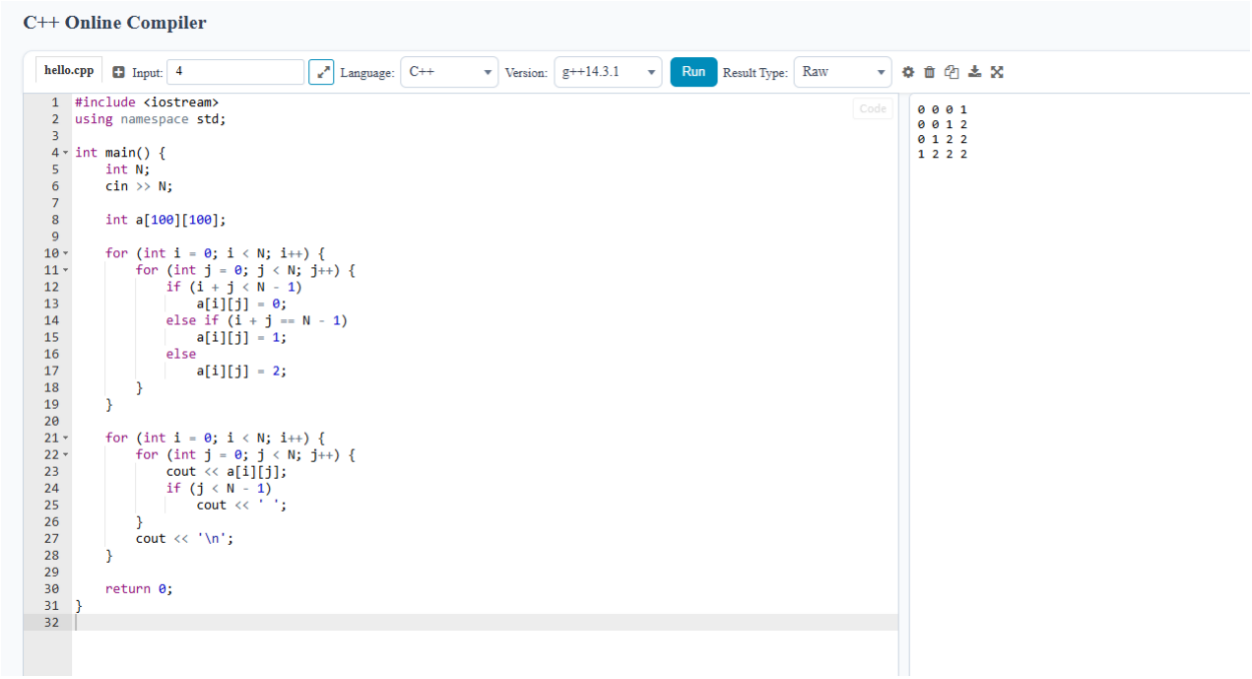
N1

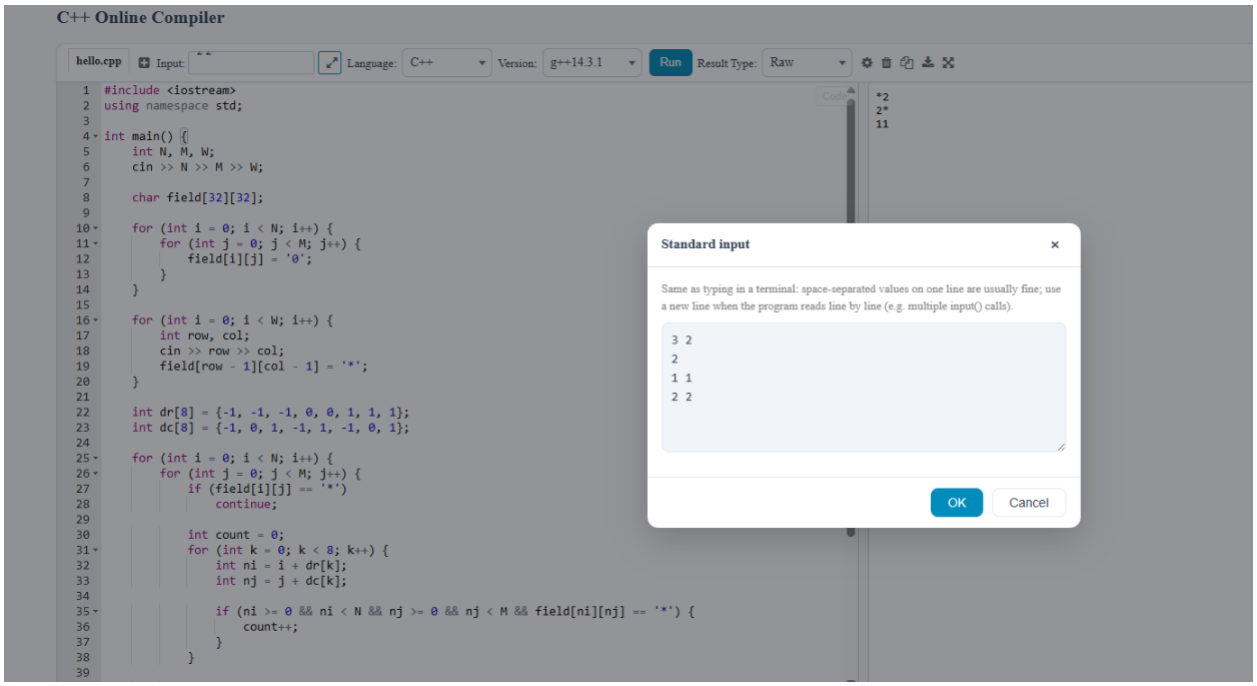



```
#include <iostream>
using namespace std;

int main() {
    int N;
    cin >> N;
    int a[100][100];
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (i + j < N - 1)
                a[i][j] = 0;
            else if (i + j == N - 1)
                a[i][j] = 1;
            else
                a[i][j] = 2;
        }
    }

    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            cout << a[i][j];
            if (j < N - 1)
                cout << ' ';
        }
        cout << '\n';
    }
    return 0;
}
```

N2



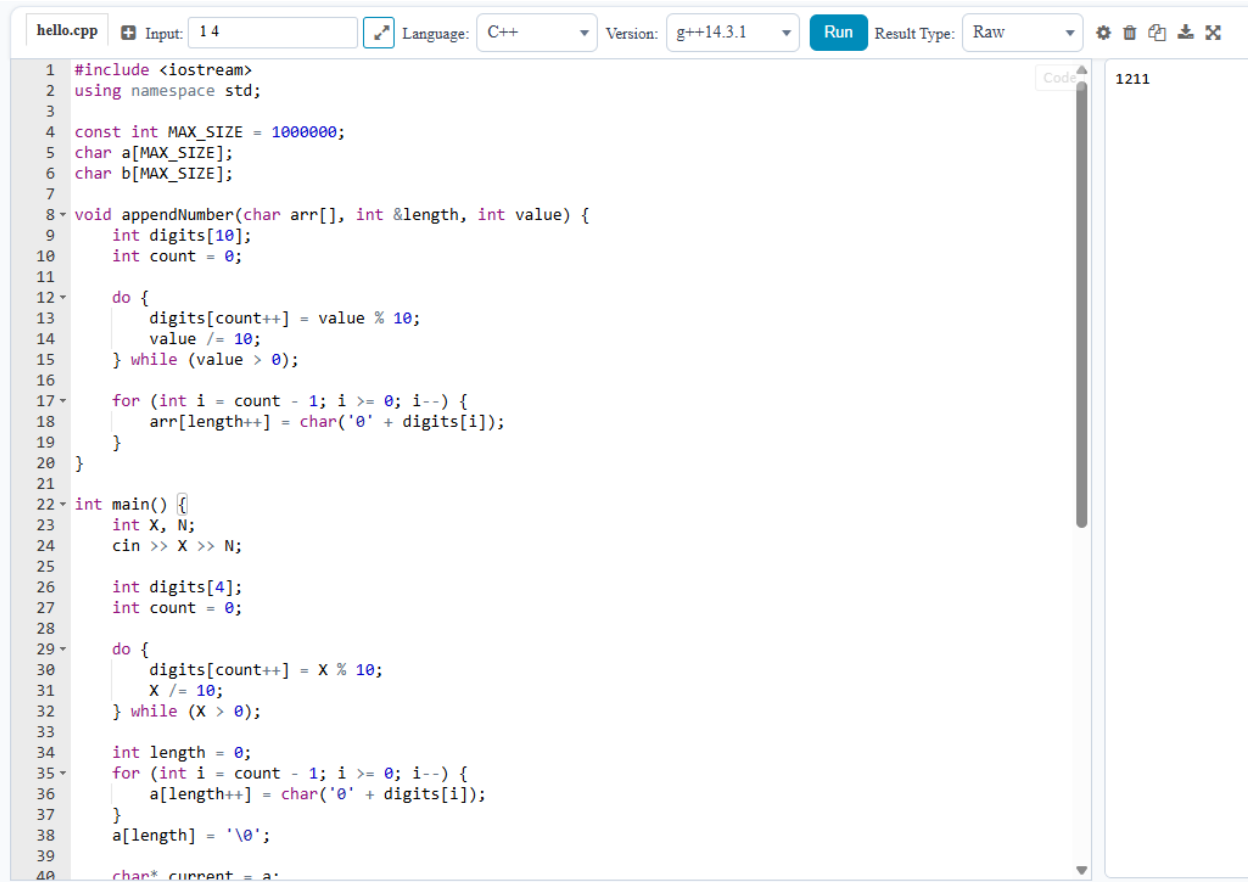

```
#include <iostream>
using namespace std;

int main() {
    int N, M, W;
    cin >> N >> M >> W;
    char field[32][32];
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < M; j++) {
            field[i][j] = '0';
        }
    }

    for (int i = 0; i < W; i++) {
        int row, col;
        cin >> row >> col;
        field[row - 1][col - 1] = '*';
    }
    int dr[8] = {-1, -1, -1, 0, 0, 1, 1, 1};
    int dc[8] = {-1, 0, 1, -1, 1, -1, 0, 1};
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < M; j++) {
            if (field[i][j] == '*')
                continue;
            int count = 0;
            for (int k = 0; k < 8; k++) {
                int ni = i + dr[k];
                int nj = j + dc[k];
                if (ni >= 0 && ni < N && nj >= 0 && nj < M && field[ni][nj] == '*') {
                    count++;
```

```
            }
        }
        field[i][j] = char('0' + count);
    }
  }
  for (int i = 0; i < N; i++) {
    for (int j = 0; j < M; j++) {
        cout << field[i][j];
    }
    cout << '\n';
  }
  return 0;
}
```

N3




```
#include <iostream>
using namespace std;

const int MAX_SIZE = 1000000;
char a[MAX_SIZE];
char b[MAX_SIZE];

void appendNumber(char arr[], int &length, int value) {
    int digits[10];
    int count = 0;

    do {
        digits[count++] = value % 10;
        value /= 10;
    } while (value > 0);

    for (int i = count - 1; i >= 0; i--) {
        arr[length++] = char('0' + digits[i]);
    }
}

int main() {
    int X, N;
    cin >> X >> N;

    int digits[4];
    int count = 0;

    do {
        digits[count++] = X % 10;
        X /= 10;
    } while (X > 0);

    int length = 0;
    for (int i = count - 1; i >= 0; i--) {
        a[length++] = char('0' + digits[i]);
    }
    a[length] = '\0';
```

```
    do {
        digits[count++] = value % 10;
        value /= 10;
    } while (value > 0);
    for (int i = count - 1; i >= 0; i--) {
        arr[length++] = char('0' + digits[i]);
    }
}
int main() {
    int X, N;
    cin >> X >> N;
    int digits[4];
    int count = 0;
    do {
        digits[count++] = X % 10;
        X /= 10;
    } while (X > 0);
    int length = 0;
    for (int i = count - 1; i >= 0; i--) {
        a[length++] = char('0' + digits[i]);
    }
    a[length] = '\0';
    char* current = a;
    char* next = b;
    for (int step = 1; step < N; step++) {
        int nextLength = 0;
        int i = 0;
        while (current[i] != '\0') {
            char digit = current[i];
            int run = 0;
            while (current[i] == digit) {
                run++;
                i++;
            }
            appendNumber(next, nextLength, run);
            next[nextLength++] = digit;
        }
        next[nextLength] = '\0';
        char* temp = current;
        current = next;
        next = temp;
    }
    cout << current << '\n';
    return 0;
}
```